

Air Quality Monitoring

Measuring the performance of AeroAlga

1. PM2.5 / PM10 Sensor

Example: PMS5003 / SDS011

- Measures the concentration of particulate matter in the air.
- Uses **light scattering**: a laser/light source illuminates particles passing through the sensing chamber.
- The scattered light is detected by a photodetector and converted into particle concentration.
- **Two sensors are placed at the inlet and outlet** of the reactor.
- Comparing the two readings allows us to calculate particulate removal efficiency.

Measurement: PM2.5 and PM10 concentration → $\mu\text{g}/\text{m}^3$

AeroAlga role: Polluted air → PM sensor → Algae reactor → PM sensor → Cleaner air

2. CO₂ Sensor

Example: MH-Z19 / MH-Z19B

- Uses **NDIR (Non-Dispersive Infrared) technology**.
- CO₂ absorbs infrared radiation at a specific wavelength.
- The sensor measures how much infrared light is absorbed to determine CO₂ concentration.
- One sensor measures CO₂ at the inlet and another at the outlet.
- A decrease in CO₂ indicates CO₂ uptake by the algae.

Measurement: CO₂ concentration → ppm

AeroAlga role: Measures the effectiveness of biological CO₂ fixation.

3. O₂ Sensor

- Measures oxygen concentration in the air.
- Can be used at the outlet to observe oxygen production associated with photosynthesis.
- Optional because PM and CO₂ measurements provide the primary evidence of air-treatment performance.
- Included only if budget and sensor availability permit.

Algae Culture Health Monitoring

Monitoring the biological conditions inside the reactor

4. pH Sensor

- Measures the acidity/alkalinity of the algae culture.
- Uses a **pH-sensitive electrode** that generates an electrical potential related to hydrogen-ion concentration.
- The signal-conditioning circuit converts this into a pH value.
- Helps determine whether the culture is maintained within a suitable range for algae growth.

Measurement: pH scale → 0–14

AeroAlga role: Ensures the biological process remains within healthy operating conditions.

5. Waterproof Temperature Sensor

Example: DS18B20

- Uses a temperature-sensitive semiconductor element to measure temperature.
- The waterproof probe can be placed directly inside the culture.
- Provides a digital temperature reading to the controller.
- Temperature affects algae metabolism, growth and photosynthetic activity.

Measurement: Temperature → °C

AeroAlga role: Monitors and records the culture temperature during operation.

6. Dissolved Oxygen (DO) Sensor

- Measures the amount of oxygen dissolved in the algae culture.
- Depending on the sensor type, measurement can be based on **electrochemical or optical detection**.
- DO provides an indication of oxygen production/consumption and biological activity.
- Useful for evaluating the health and activity of the culture.

Measurement: Dissolved oxygen → mg/L or % saturation

AeroAlga role: Helps monitor biological activity inside the photobioreactor.

7. Turbidity Sensor

- Uses a light source and photodetector to measure how much light is scattered by particles in the water.
- As algae biomass increases, the culture generally becomes more turbid.
- Therefore, turbidity can be used as a **proxy for algae growth/biomass**.
- It should ideally be calibrated against actual biomass measurements.

Measurement: Turbidity → NTU / sensor output

AeroAlga role: Provides continuous indication of changes in algae concentration.

Engineering Measurements & Data Acquisition

Turning sensor readings into measurable performance

8. Airflow Sensor / Flow Meter

- Measures the rate at which air passes through the reactor.
- Usually works by detecting air movement through a sensing element or calibrated flow path.
- Allows us to determine how much air is being treated.
- Also helps estimate **air residence time** inside the reactor.

Measurement: Airflow → L/min or m³/h

Why it matters: A pollution-removal percentage is much more meaningful when we also know the amount of air being processed.

9. Light Sensor

Example: BH1750 or equivalent

- Measures the intensity of light reaching the reactor.
- Important because algae require light for photosynthesis.
- Allows us to monitor whether illumination remains consistent during experiments.

Measurement: Light intensity → lux

AeroAlga role: Records the lighting conditions under which CO₂ fixation and algae growth occur.

Data Acquisition System

ESP32 Microcontroller

Sensors → ESP32 → Data Processing → Storage / Display

The ESP32 acts as the central data-acquisition unit:

1. Collects readings from the air-quality and culture sensors.
2. Records measurements at regular intervals.
3. Stores experimental data on an SD card.
4. Displays important values locally or sends them to a computer/dashboard.
5. Calculates inlet-vs-outlet performance.

Key Performance Calculations

PM Removal Efficiency

$$\eta = \frac{PM_{in} - PM_{out}}{PM_{in}} \times 100$$

CO₂ Reduction

$$\eta = \frac{CO_{2,in} - CO_{2,out}}{CO_{2,in}} \times 100$$

Overall Concept

Polluted Air → Inlet Measurement → Algae Photobioreactor → Outlet Measurement → Performance Analysis

The instrumentation system therefore provides quantitative evidence for whether AeroAlga is actually reducing particulate matter and CO₂ while maintaining suitable conditions for algae growth.